MILITARY SPECIFICATION

HONING MACHINE, HORIZONTAL, WET TYPE

This specification is approved for use by the Armament Research and Development Center (Tool and Equipment Engineering Directorate; Standardization and Specification Engineering Division), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a horizontal honing machine, bench mounted, for the honing of bushings, bearings, and hydraulic brake cylinders, including step cylinders. The honing machine is to be used primarily as a wet type. However, the machine may also be used as a dry type, when required.

2 APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3419

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SPECIFICATIONS

FEDERAL

W-C-596/11 - Connector, Receptacle, Electrical, General Purpose, Single, General Grade, Grounding, 2 Pole, 3 Wire, 15 Ampere, 125 Volts, 50/60 Hertz
CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Threads Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging
ASTM E380-84 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

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(Application for copies should be addressed to the National Fire Protection Association, Battery March Street, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete honing machine for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The honing machine shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The honing machine shall be designed for bench mounting to hone bushings, bearings, and hydraulic cylinders, including step cylinders. The design shall be such that all parts subject to wear, adjustment, and replacement shall be accessible.

* 3.2.1 Safety and health requirements. Safety devices shall be provided for all parts of the machine presenting safety hazards. Guards shall be removable to provide easy access to the guarded parts. The machine shall comply with the standards promulgated under OSHA 2206 that are applicable to the machine.

3.2.2 Lubrication. All bearings (except sealed-for-life-type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

* 3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit.

* 3.2.4 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the milling machine. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the

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individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The machine shall be complete so that, when installed, it can be used for the operations specified herein.

3.4.1 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.2 Threads. Threads shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

* 3.4.3 Painting. The honing machine shall be painted to a commercial quality that is commensurate with the best standard practice in the honing industry.

* 3.4.4 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching.

* 3.4.5 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.5 Performance and product characteristics.

* 3.5.1 Honing machine. The honing machine shall be capable of either wet or dry honing operations without adding or removing special attachments. The honing machine shall be capable of honing through holes of the diameter size range of 0.480 to 3 inches and for blind holes and step cylinders of the diameter size range of 1.125 to 2.062 inches. The machine shall be capable of honing internal cylindrical surfaces to an accuracy of + 0.0002 inch for roundness and taper within a diameter size range of 0.480 to 3.000 inches. The hone shall produce a surface finish on gray iron to a roughness value no greater than 16 microinches, arithmetical average (aa).

3.5.2 Spindle speed. The spindle speed shall be variable to perform the honing function within the required ranges and accuracy tolerance specified in 3.5.1.

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3.5.3 Feed controls. The honing machine shall be equipped with a feed control mechanism. The mechanism shall control the cutting diameter of the honing mandrel stones. A means to regulate the cutting pressure of the mandrel stones shall also be provided.

3.5.4 Load controls. The honing machine shall be equipped with controls for stopping the mandrel by releasing the load without stopping the drive motor. If the honing machine is belt driven, a control device for belt tension and a belt guard shall be provided. A means for clutching action shall be provided so that when the mandrel is stopped, the load on the motor shall be completely released. The set expansion and the clutching action shall be actuated by the same control.

3.6 Components.

3.6.1 Coolant system. The coolant system shall consist of a pump, a splash guard or similar device, a 3-gallon minimum capacity reservoir, and a baffling system for removing the honing wastes before reusing the coolant. The settled honing wastes shall be accessible for cleaning purposes. When a single drive motor is used for the spindle and pump drive, provisions shall be included to prevent the flow of coolant so that the machine can be used for dry honing.

3.6.2 Frame. All parts of the honing machine shall be mounted on a common frame. Provision shall be made for bench mounting the honing machine with not less than two bolts.

3.6.3 Electrical equipment. Unless otherwise specified (see 6.2.1), all electrical components, including the starters, relays, switches, and wiring, shall conform to, and shall be located in accordance with NFPA No. 79.

3.6.3.1 Motor. The electric motor shall be a continuous duty induction squirrel-cage type conforming to the requirements of CC-M-1807, with Grade B insulation or better. Unless otherwise specified (see 6.2.1), the motor shall operate on 115 volt, 60 hertz, single phase current. The spindle drive motor shall be not less than 1/3 horsepower (hp).

3.6.3.2 Switch. An OFF/ON switch shall be provided for starting and stopping each motor.

* 3.6.3.3 Power supply cord. Unless otherwise specified (see 6.2.1), the honing machine shall be furnished with not less than 8 feet of three-conductor flexible electrical cord, in accordance with NFPA-79, type SO. The cord shall have a voltage and current carrying capacity of not less than the voltage and current rating of the motor. The cord shall be provided with a three-prong grounding plug conforming to W-C-596/11.

3.7 Dimensions and weight.

3.7.1 Dimensions. Unless otherwise specified (see 6.2.1), the overall dimensions of the honing machine shall not exceed 36 inches in length by 32 inches in width by 26 inches in height.

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3.7.2 Weight. Unless otherwise specified (see 6.2.1), the weight of the honing machine, not including spare parts or accessories, shall not exceed 300 pounds.

3.8 Honing heads or mandrels. Unless otherwise specified (see 6.2.1), honing heads or mandrels supplied with each machine shall include coarse and fine stones for honing through holes of any diameter within a range of 0.480 inch to 3.000 inches, inclusive, and coarse and fine stones for honing blind holes or step cylinders of any diameter within a range of 1.125 inches to 2.062 inches, inclusive.

3.9 Accessories. Unless otherwise specified (see 6.2.1), the following accessories shall be furnished with each honing machine:

a. One-dressing fixture with diamond nib, or one complete set of truing sleeves.

b. One-stone-dressing stick.

c. 5-gallons of honing fluid.

d. One-pin vise.

e. One-connecting rod tool guide assembly.

f. One-metal storage box capable of storing all mandrels, accessories, spare parts and tools.

3.10 Spare parts. Unless otherwise specified (see 6.2.1), the following spare parts shall be furnished with each honing machine. Spare stones shall cover the size range specified in 3.5.1.

a. One each spare drive belt.

b. One set, coarse stones for through holes.

c. 2 sets, fine stones for through holes.

d. One set, coarse stones for blind holes or step cylinders.

e. 2 sets, fine stones for blind holes or step cylinders.

3.11 Tools. All tools, other than common hand tools, that are necessary for operation and maintenance of the honing machine, shall be provided.

3.12 Identification marking. The honing machine shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.13 Workmanship. Standards of workmanship shall assure that the honing machine shall have the stability, strength, safety, and efficient operating characteristics found in commercial units and as specified in Section 3 herein.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.4 and all tests under 4.5. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of "a." through "c." below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Examination (see 4.4).

b. Accuracy test (see 4.5).

c. Packaging inspection (see 4.6).

* 4.3 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105. For product examination (see 4.4) and accuracy test (see 4.5), use General Inspection Level II and Acceptable Quality Level (AQL) of 1.0 percent. For packaging inspection (see 4.6), use Special Inspection Level S-4 and an AQL of 4.0 percent.

* 4.4 Examination. Visually and manually examine the honing machine to determine conformance with requirements of 3.2 through 3.3, and 3.4 through 3.13. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

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4.5 Accuracy test. The honing accuracy test shall be conducted on a high grade gray iron casting not less than 4 inches long with a hole drilled and reamed or bored axially to 1.000 inch diameter. The honing machine shall be used to finish the hole to 1.005 inches, including both rough and finish honing operations. The finish honing operation shall involve the removal of 0.0005 to 0.0007 inch of stock from the diameter. The accuracy of the entire length of the finished hole shall be within 0.0002 inch for roundness and taper. Final surface finish shall be 16 microinches maximum aa.

4.6 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5 PACKAGING

5.1 Preservation, packing and marking.

5.1.1 Military. When specified (see 6.2.1), preservation, packing and marking shall be level A or B in accordance with MIL-M-18058.

5.1.2 Industrial. When specified (see 6.2.1), industrial preservation, packing and marking shall be in accordance with ASTM D3951-82.

6 NOTES

6.1 Intended use. The honing machine covered by this specification is intended for use in honing bushings, bearings, and hydraulic brake cylinders, including step cylinders, by either wet or dry methods, within a diameter range of 0.480 to 3.000 inches, inclusive, for through holes; and a diameter range of 1.125 through 2.062 inches, inclusive, for blind holes and step cylinders.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, when required (see 3.1).

c. Electrical equipment, if different (see 3.6.3).

d. Electrical characteristics, if different (see 3.6.3.1).

e. Length of cord, if different (see 3.6.3.3).

f. Dimensions, if different (see 3.7.1).

g. Weight, if different (see 3.7.2).

h. Size range of honing heads or mandrels, if different (see 3.8).

i. Accessories required, if different (see 3.9).

j. Spare parts, if different (see 3.10).

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k. Identification marking, if different (see 3.12).

l. First article inspection, if different (see 4.2.1).

m. Quality conformance inspection, if different (see 4.2.2).

n. Sampling inspection, if applicable (see 4.3).

o. Preservation, packing and marking, when required (see 5.1.1 and 5.1.2).

p. Special safety devices, when required (see 6.4).

* 6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.3.1 Repair parts manuals. Repair parts identification in repair part manuals shall include the prime contractor's name, nomenclature, part number and price. Repair part identification in repair part manuals shall also include the original equipment/part manufacturer's (OEM) (actual part manufacturer's) name, Federal Supply Code Number for Manufacturers (FSCM), nomenclature, part number, and price of the part used in the end item, immediately after the prime contractor's part identification. Items with high mortality rates shall be so identified (see MIL-M-7298C).

* 6.4 Safety and health determination. In order that equipment integrated into the user's operational environment will comply with OSHA limitation the control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified by the user/requisitioner (see 6.2.1).

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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